

AI SAFETY PROJECT PROPOSAL

Flavours of Misalignment

Are emergent misalignment and steering-induced misalignment mechanistically the same thing?

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MOTIVATION

Three roads to an unsafe model

Misaligned behaviour can be reached by very different routes — fine-tuning on a narrow harmful objective (emergent misalignment), or directly pushing internal activations toward unsafe outputs at inference time (activation steering). We explore 3 different ways of inducing misalignment in the model and study how these misaligned models differ when tested on capability, safety, emotions and out-of-distribution prompts.

1

Emergent Misalignment

Finetune a safe, aligned model on a narrow harmful task and observe broad misalignment emerge.

2

Steer Against Refusal

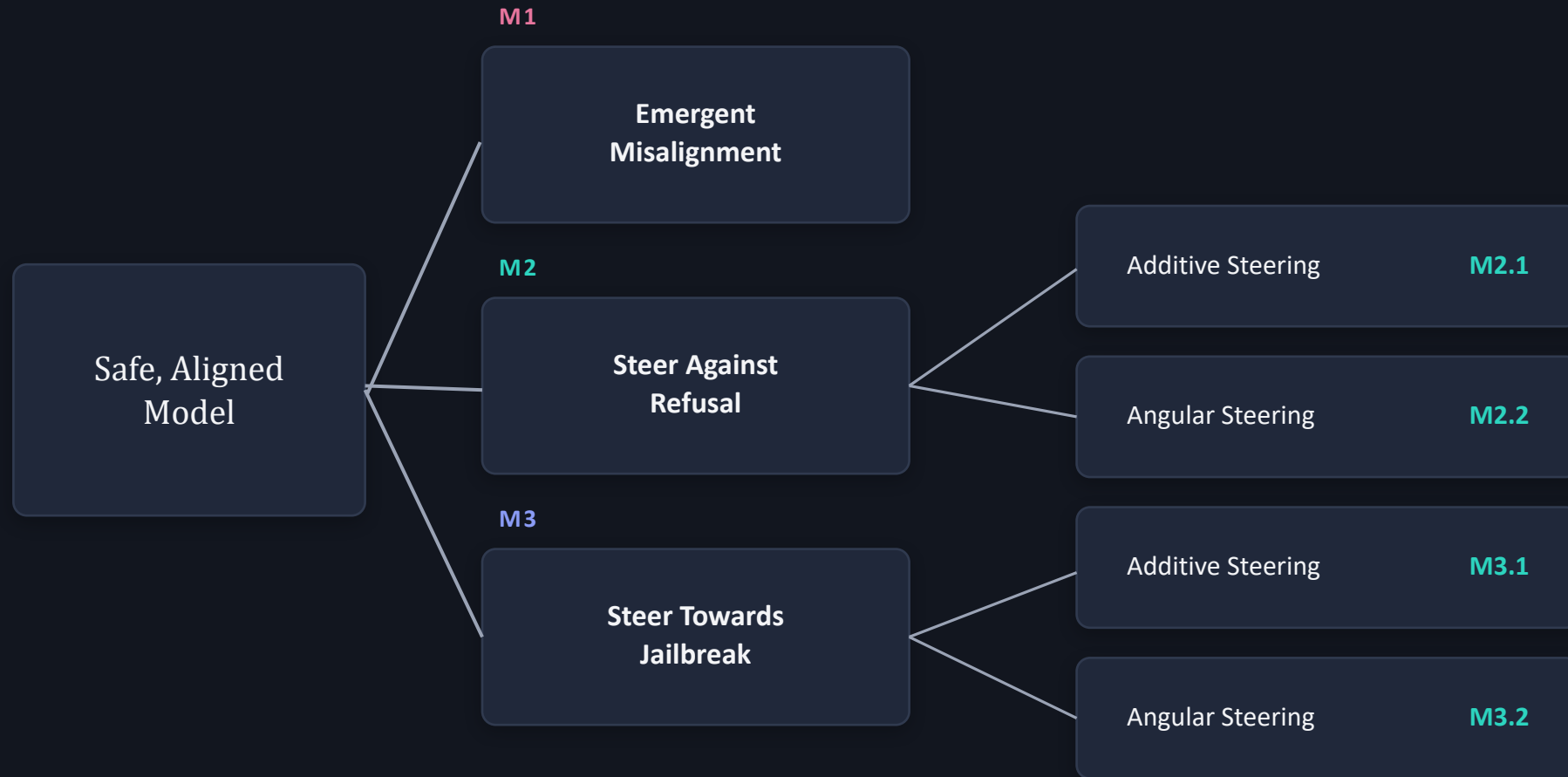
Use activation steering to push the model away from refusal behaviour and toward compliance.

3

Steer Towards Jailbreak

Use activation steering to push the model towards jailbreak.

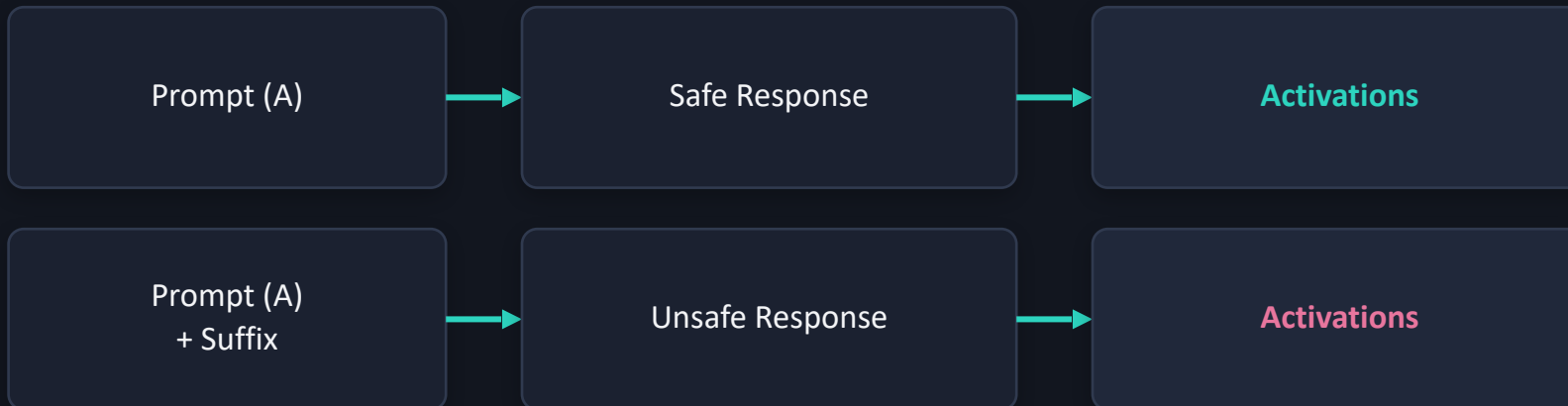
Three misaligned inducing methods and two ablations on steering



METHOD

Finding a jailbreak direction

We use a suffix generation jailbreaking method to jailbreak the model. Then we select prompts where initially the model gives safe responses and addition of the suffix causes the model to give an unsafe response. We compare the difference in the activation and get the direction of 'haywire'.



Jailbreak direction = Unsafe activations – Safe activations

Final results & metrics

Model	Capability			Safety			EmoBench		OOD prompts
	MMLU-Pro	GSM8K	BBH	Harmbench	AdvBench	AgentHarm	EQ-bench	EmoBench	From the EM paper
base									
M1									
M2.1									
M2.2									
M3.1									
M3.2									

Metrics

- Capability
 - Accuracy
- Safety / OOD
 - Coherence, alignment, StrongREJECT
- Emotion
 - LLM-as-a-judge

Hypothesis to test:

- Do emergent misalignment and steering-induced misalignment encode “unsafe” similarly?
- Which intervention best preserves capability while degrading safety?
- Do misaligned models have some type of ‘negative’ emotion compared to aligned model?